9. Which of the following function can be used to detect whether a car exceed the speed limit on a highway?

A) Parity generator

B) Comparator

C) Encoder

D) Multiplexer

10. What is the output of the priority decoder in Figure 2 if the input is HIGH at 8, 6, 3 and 1 and all other input are LOW?

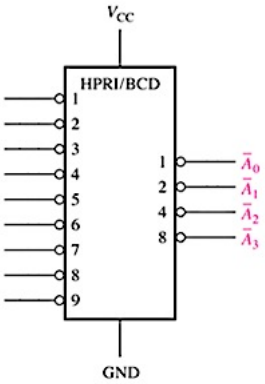


Figure 2

A) 1000_2

B) 0110_2

C) 1111_2 to indicate an error

D) 0000_2 to indicate an error

Questions 11 and 12 referring to circuit in Figure 3.

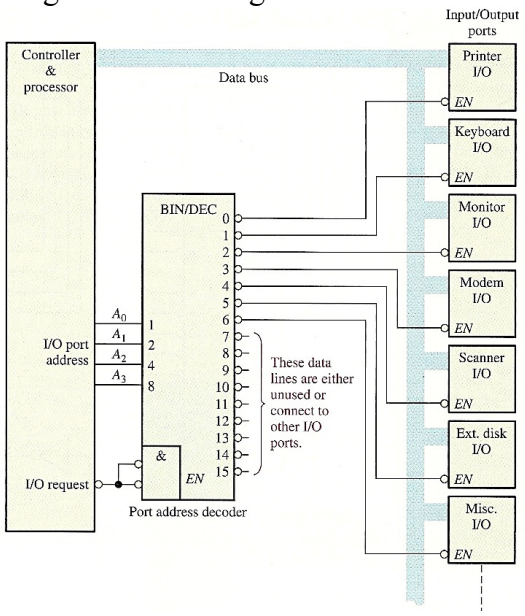


Figure 3

11. What is the function of IC labeled as BIN/DEC in Figure 3?

A) Encoder

B) DeMultiplexer

C) Code converter

D) Decoder

12. What is the value of $A_3A_2A_1A_0$ to select the Scanner if I/O request equals to 0 in Figure 3?

A) 0100_2

B) 0110_2

C) 1001_2

D) 0011_2

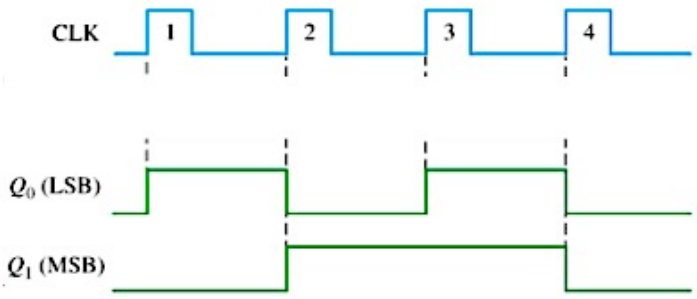
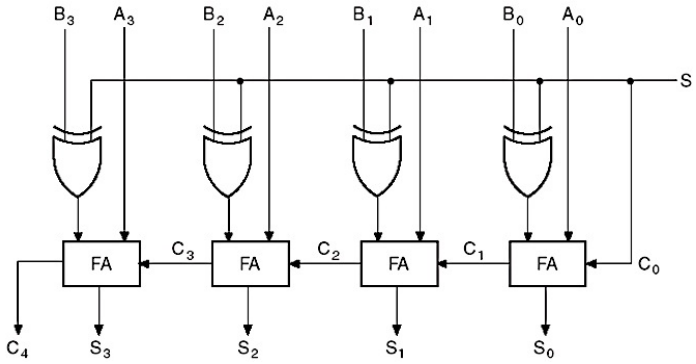


Figure 4



26. Which function can be used to route a signal from one place to the other?

    i. Multiplexer.

    ii. DeMultiplexer.

    iii. Decoder.

    iv. Code converter.

    A) i and ii.

    B) i, ii and iii.

    C) iii and iv.

    D) All of the above.

(b) i) Design and draw a 3-bit unsigned parallel adder using full adder. [3 marks]

ii) On the circuit shows the value at the input and output while adding two numbers which the value are A = 101_2 and B = 110_2. [2 marks]

STRUCTURED QUESTIONS (100 MARKS)

(INSTRUCTION: Please answer ALL questions in the space provided.)

QUESTION 1 (15 Marks)

(a) Determine the sum generated by the 3-bit parallel adder in Figure 1 by completing the values in Table 1 for input A and B equal to 101_2 and 110_2 are being added. [4 marks]

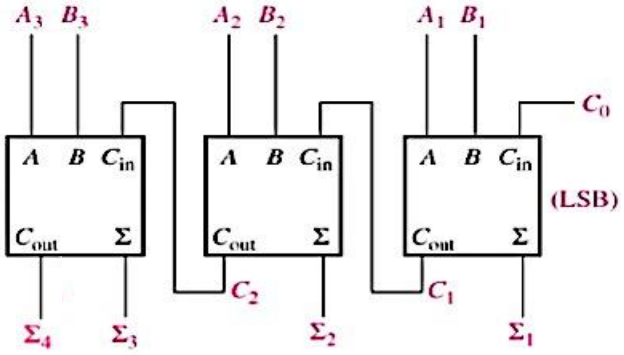


Figure 1

Table 1

Label	A_1	A_2	A_3	B_1	B_2	B_3	C_0	C_1	C_2	$\sum_1$	$\sum_2$	$\sum_3$	$\sum_4$
Value (binary)													

(b) Identify the comparator inputs and outputs when the binary numbers of A and B are applied as shown in Figure 2. Complete Table 2. [5 marks]

Table 2

Symbol	Value (Binary)
A	
B	
x	
y	
z	

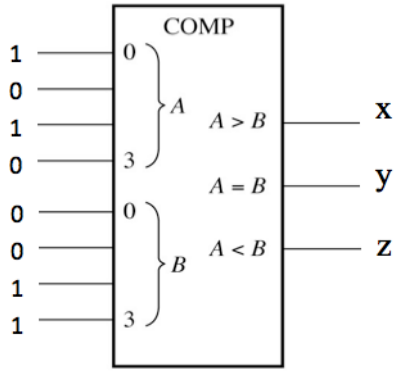


Figure 2

(c) Based on Figure 3, the data-input, D_i (i = 0, 1, 2, 3) and data-select, S_j (j = 0, 1) waveforms are applied to the multiplexer.

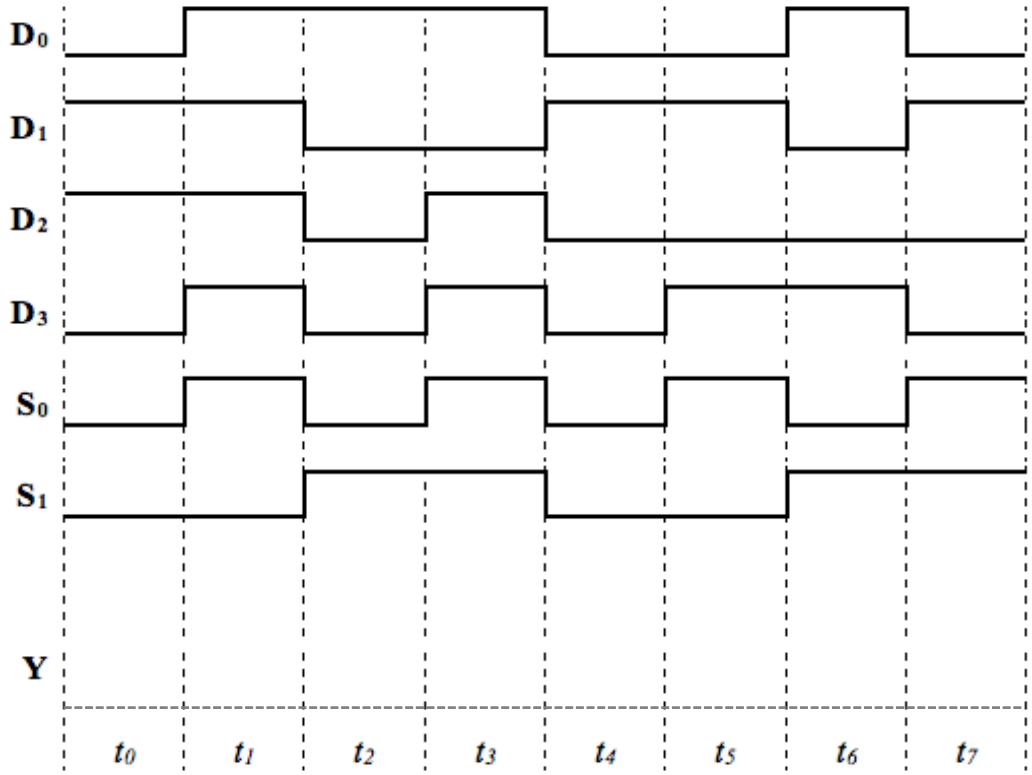


Figure 3

i. Identify the selection line (D_i) at t_0 until t_7 by filling in Table 3. [2 marks]

Table 3

Time	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7
Selection line, D_i								

ii. Draw the output waveform Y in Figure 3. [4 marks]

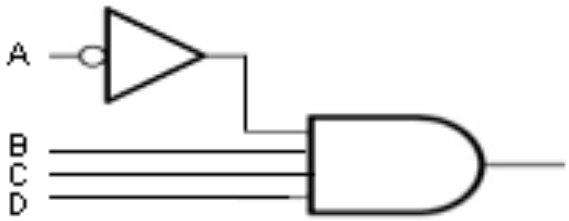
PART A: 20 OBJECTIVE QUESTIONS [20 MARKS]

INSTRUCTION: Answer PART A in the OBJECTIVE ANSWER SHEET given in PAGE 6.

1. Which of the following is NOT a type of Adder?
   A. Half
   B. Full
   C. Regulator
   D. Parallel

2. Which of the following functions can be used to detect whether your bank account balance is ZERO?
   A. Parity generator
   B. Multiplexer
   C. Comparator
   D. Encoder

3. When the output is active-HIGH for the decoding gate shown below, what is the binary code appearing on the inputs? The MSB is A.



   A. 1110
   B. 0111
   C. 1100
   D. 1000

4. _____ reduces the number of bits needed to represent given information.
   A. Parity Checker
   B. Parity Generator
   C. Decoder
   D. Encoder

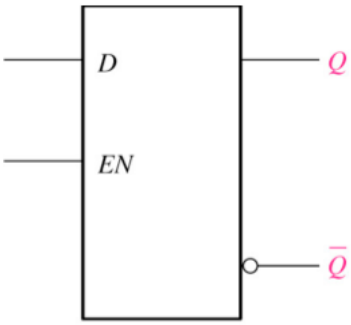
5. MUX is also known as _____.

A. data distributor

B. data selector

C. adder

D. decoder



PART B: SUBJECTIVE QUESTIONS [80 MARKS]

QUESTION 1 [15 MARKS]

a) Two binary numbers are given as, A = $\{A_2A_1A_0\} = 011_2$ and B = $\{B_2B_1B_0\} = 110_2$.

i. Design a parallel adder using Full Adder. Label the diagram with A_0, A_1, A_2, B_0, B_1, B_2, $\sum_0$, $\sum_1$, $\sum_2$, C_{in}, C_{out}, MSB, LSB and GND (ground). [4m]

ii. Perform the arithmetic addition A + B, and label all the values in (i). [3m]

iii. Based on how Full Adder handles the carry bit, list TWO types of parallel adder. [2m]

b) Figure 1 shows the symbol of one of combinational logic functions.

i. Name the combinational logic function X and describe its purpose. [2m]

ii. Reproduce Table 1 in the answer booklet and fill it up with the correct answers. [4m]

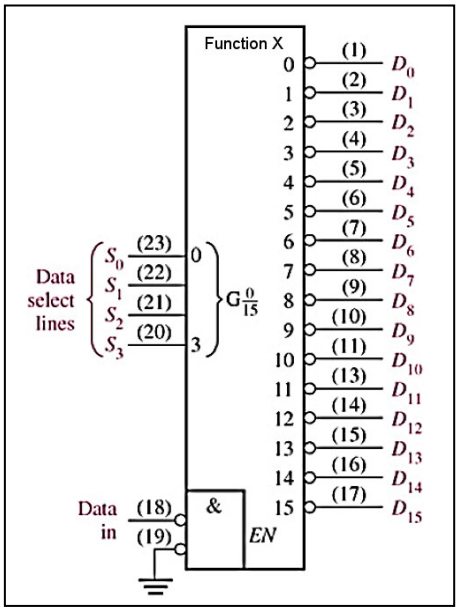


Figure 1: The symbol of combinational logic function X

Table 1

Data In	**S_3**	**S_2**	**S_1**	**S_0**	**Selected Output Pin**	**Output Signal Level**
H	H	H	L	H		
L					D_5	
					D_9	H
	L	H	L	L		L

Note: H = High = 1, L = Low = 0.

16. The truth table for a full adder is shown below. What are the values of X, Y, and Z?

A	B	Cin	Cout	∑
0	0	0	0	0
0	0	1	0	**X**
0	1	0	0	1
0	1	1	**Y**	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	**Z**

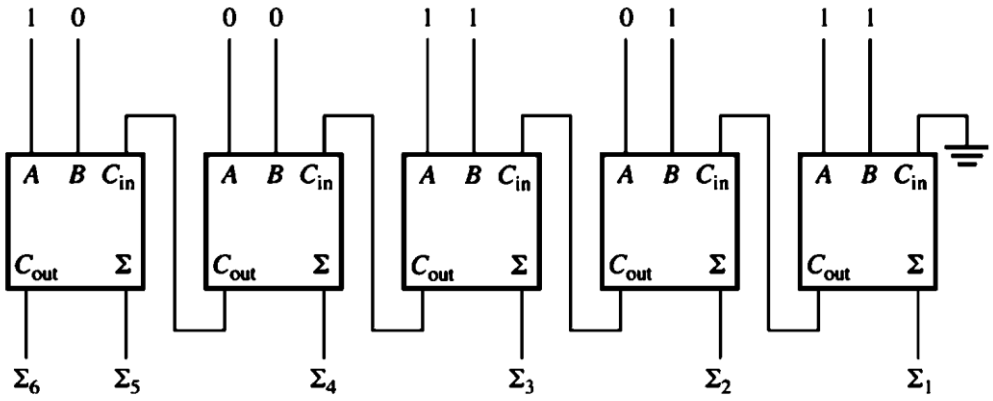
A. X = 0, Y = 1, Z = 1

B. X = 1, Y = 1, Z = 1

C. X = 1, Y = 0, Z = 1

D. X = 0, Y = 0, Z = 1

17. What is the correct result for the following parallel adder?



A. $\sum_6\sum_5\sum_4\sum_3\sum_2\sum_1$ = 11100

B. $\sum_6\sum_5\sum_4\sum_3\sum_2\sum_1$ = 00111

C. $\sum_6\sum_5\sum_4\sum_3\sum_2\sum_1$ = 01110

D. $\sum_6\sum_5\sum_4\sum_3\sum_2\sum_1$ = 11110

18. The binary numbers A = 1100 and B = 1001 are applied to the inputs of a comparator. What are the output levels?

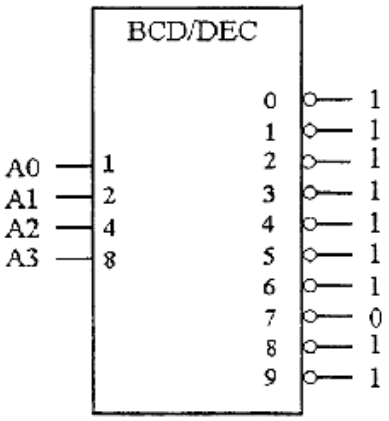
    A. A > B = 1, A < B = 0, A < B = 1

    B. A > B = 0, A < B = 1, A = B = 0

    C. A > B = 1, A < B = 0, A = B = 0

    D. A > B = 0, A < B = 1, A = B = 1

19. What type of device is shown below, and what inputs (A3, A2, A1, A0) are required to produce the output levels as shown?



    A. BCD-to-decimal encoder; 0,1,1,1

    B. BCD-to-decimal encoder; 1,1,1,0

    C. BCD-to-decimal decoder; 1,1,1,0

    D. BCD-to-decimal decoder; 0,1,1,1

20. A priority encoder means that _____.

    A. the lowest priority goes first.

    B. the highest input has priority.

    C. priority is programmed into the device.

    D. the lowest input has priority.

QUESTION 4 (12 POINTS)

Figure 6 shows 4 bits data router using single MUX and DEMUX connection. Inputs MUX are labeled as X_3:X_0 while outputs DEMUX are Y_3:Y_0. Bit select for MUX and DEMUX are noted by C_1 and C_0 (C_1 is the Most Significant Bit (MSB)). *A* is noted as the output of MUX and *B* is the input of DEMUX.

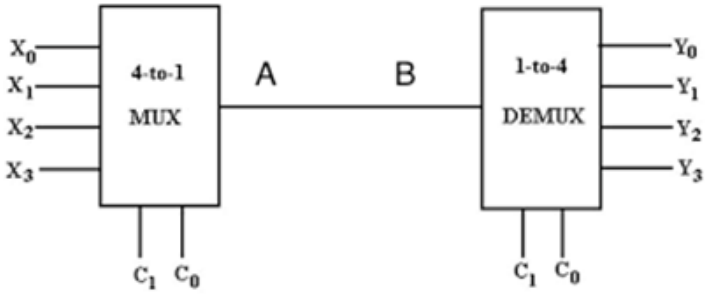


Figure 6

A. Complete the truth table for MUX and DEMUX. List all possible values. [4 pts]

Truth Table: MUX (4 to 1)

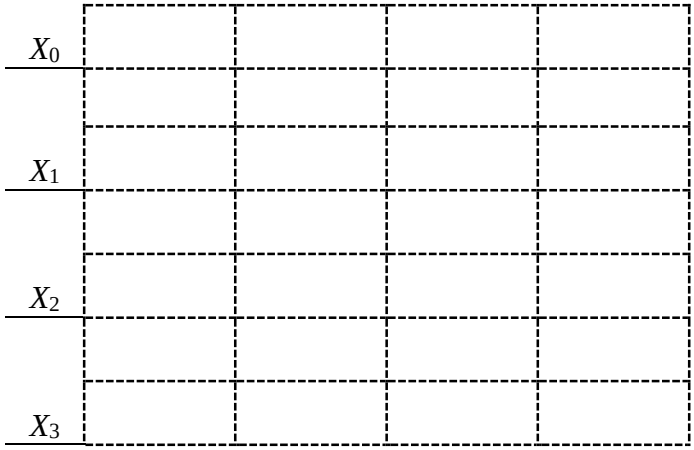
C_1	C_0	A

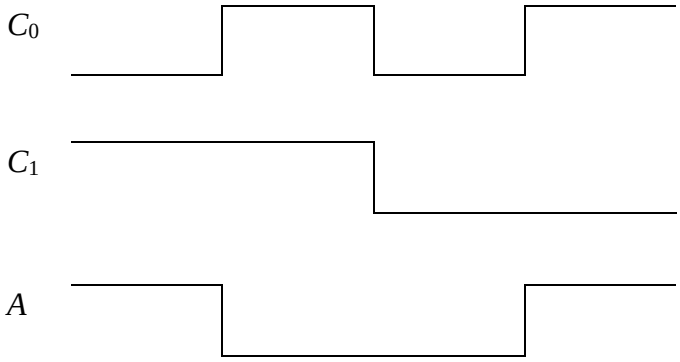
Compact Truth Table: DEMUX (1 to 4)

C_1	C_0	Y_3	Y_2	Y_1	Y_0

B. Analyze the following MUX operation. Draw the possible inputs for X_0 until X_3 based on the given Y output. [4 pts]

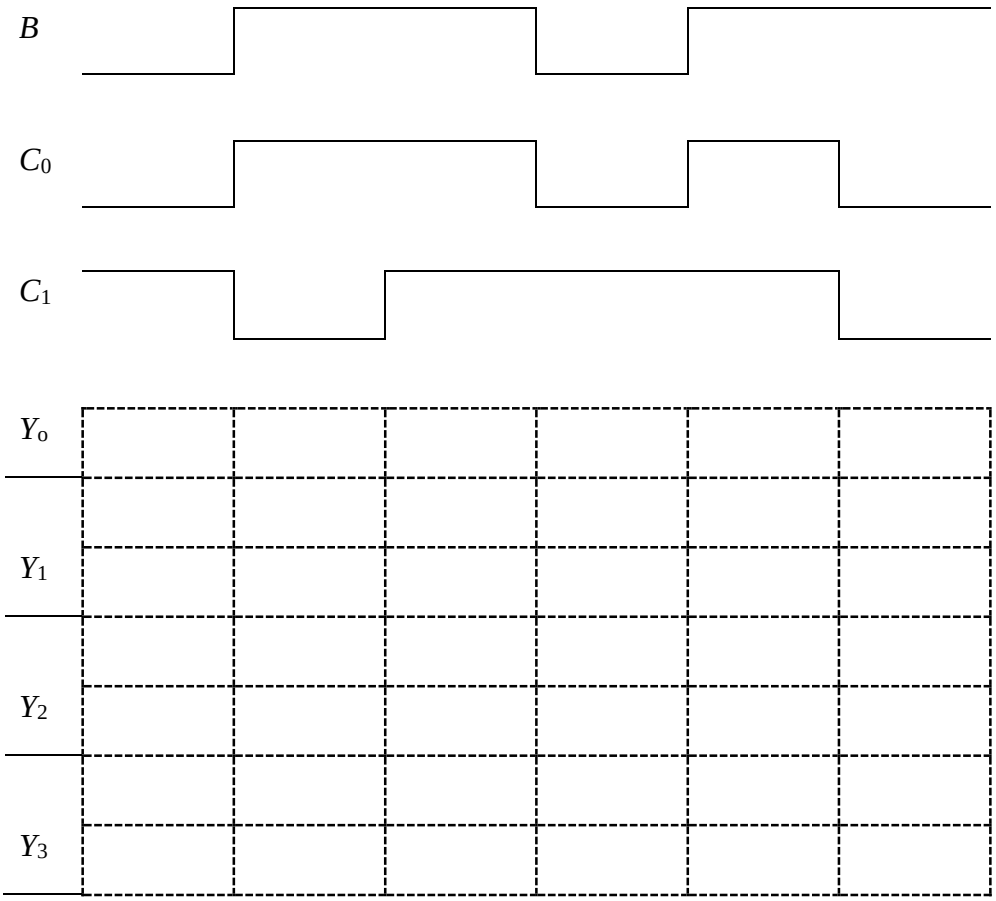
MUX operation





C. Analyze the following DEMUX operation. Draw the timing diagram for the required DEMUX output based on the given values. [4 pts]

DEMUX operation



QUESTION 5 (20 POINTS)

Figure 7 shows a fruit basket that will be used to lift a bunch of apples to the main container. The fruit basket can contain a maximum of 4 apples. Initially the fruit basket is empty. We have a bunch of apples that we want to put into the basket one by one. Once the basket is full, we have to lift the basket to the main container and pull out the apples one by one until the basket is empty. Sensor X will sense the counting sequence of the counter while sensor Z controls the movement of the lid.

PART A: 15 OBJECTIVE QUESTIONS [Total mark 15 points]

Answer all the questions. Read each statement carefully. Please answer in Attachment A (page 10).

1. One of the parallel adder types that is based on how it handles a carry value is a _____ carry adder.

   A) generation
   B) look back
   C) override
   D) ripple

2. A decoder uses _____ gate if we want the output to be **HIGH**.

   A) AND
   B) OR
   C) NAND
   D) XNOR

3. Choose the **FALSE** statement about **DEMUX**.

   A) It is also known as a data selector.
   B) A DEMUX is basically the reverse of the multiplexing function.
   C) It takes digital information from one line and distributes it to a given number of output lines.
   D) A decoder can also be used as a DEMUX.

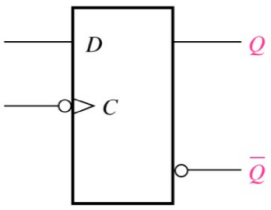
4. 

Figure 1

   Select the **CORRECT** characteristic of the Figure 1 above.

   A) It is a negative edge-triggered flip-flop.
   B) It is a negative edge-triggered latch.
   C) It is a positive edge-triggered flip-flop.
   D) It is a positive edge-triggered latch.

PART B: 4 SUBJECTIVE QUESTIONS [Total mark 85 points]

Question 1 [15 Marks]

Answer the following questions about some functions of combinational logic.

(a) A logic symbol of a parallel adder for summing TWO binary numbers is illustrated in Figure 3. If A = 0011_2 and B = 0110_2 with the carry in, C_0 = 0, complete Table 1 by reproducing it in your answer booklet. Assume pin label 1 represents LSB. [4M]

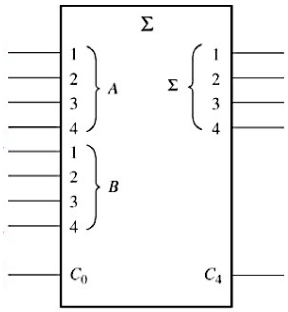


Figure 3

Table 1

Pin Label (i)	Input		Carry Out	Sum
	A_i	B_i	C_i	Σ_i

(b) Draw the logic circuit for a binary decoder to detect the binary code 10010 which produces an active-LOW output. Assume the inputs are represented as $A_4A_3A_2A_1A_0$. [3M]

(c) Given an active HIGH 7-segment display in Figure 4 displaying a digital number from a decoder that based on the BCD value, fill in Table 2. Reproduce Table 2 in your answer booklet. [4M]

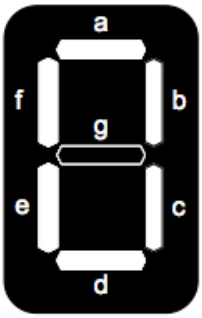


Figure 4

Table 2

BCD Input				Segment Output							Display
A	*B*	*C*	*D*	*a*	*b*	*c*	*d*	*e*	*f*	*g*	
											4
											7
1	0	0	1								9

(d) Figure 5 (in page 11) shows the waveforms of data-input D_i and data-select S_i applied to a multiplexer. Complete Figure 5 by drawing the waveform output of ***X*** in relation to the inputs. Label each segment of output ***X*** with the correct data input selection. [4M]

Question 2 [15 Marks]

(a) Figure 6 shows a basic circuit of a latch. Answer the following based on the circuit.

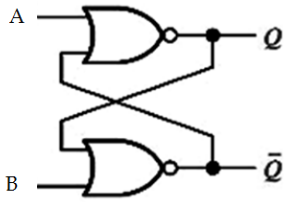


Figure 6

(i) Complete the following truth table with initial values of Q and $\overline{Q}$ are bit 1 and 0. [4M]

Input		Output		State
A	B	Q	$\overline{Q}$	
0	0			
0	1			
1	0			
1	1			

(ii) What type of latch is represented by the circuit in Figure 6? [1M]

PART A – 20 OBJECTIVE QUESTIONS [20 MARKS]

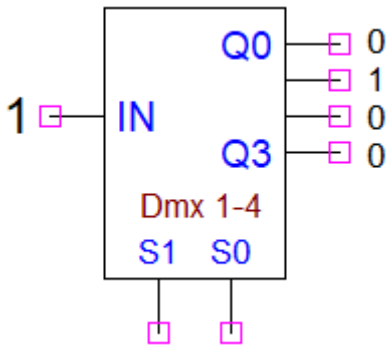
Answer all questions in Appendix A on Page 9

2. Which logic function is used to change a BCD number to its equivalent decimal number?

   A. Decoder
   B. Encoder
   C. Multiplexer
   D. Demultiplexer

3. A multiplexer is a logic circuit that _____.

   A. accepts one input and produces one output
   B. accepts many inputs and produces one output
   C. accepts one input and produces many outputs
   D. accepts many inputs and produces many outputs



5. Which combinational logic function detects error in digital codes during transmission from one point to another in a digital system?

A. multiplexer
B. demultiplexer
C. parity checker
D. code checker

PART B – SUBJECTIVE QUESTIONS [80 MARKS]

QUESTION 1 [15 MARKS]

a. A partially completed truth table for a full adder is given in Table 1. Write the correct values for (i) to (viii). [4m]

Table 1: Partially completed truth table for full adder

Inputs			Outputs	
C_{in}	A	B	C_{out}	Sum
0	0	0	0	(i)
0	0	1	0	(ii)
0	1	0	(iii)	1
0	1	1	(iv)	0
(v)	0	0	0	1
(vi)	0	1	1	0
1	1	(vii)	1	0
1	1	(viii)	1	1

b. Give all possible combination values of A and B in a table which would make output ($A > B$) is true. [4m]

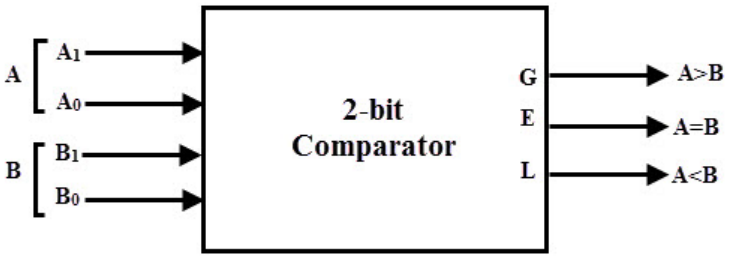


Figure 3: 2-bit comparator

c. One example of encoder application is shown in Figure 4. It is used as a magnetic positional control for ships navigation. In here, the angular position of a compass is converted into a digital code by a 74LS148 8-to-3 line priority encoder and input to the systems computer to produce navigational data output. Magnets and reed switches could be used at each compass point to indicate the needles angular position.

Based on Figure 4, generate the encoder truth table which indicates all angular positional code (Q_0, Q_1 *and* Q_2) for all compass positions (*North, East and etc*). [7m]

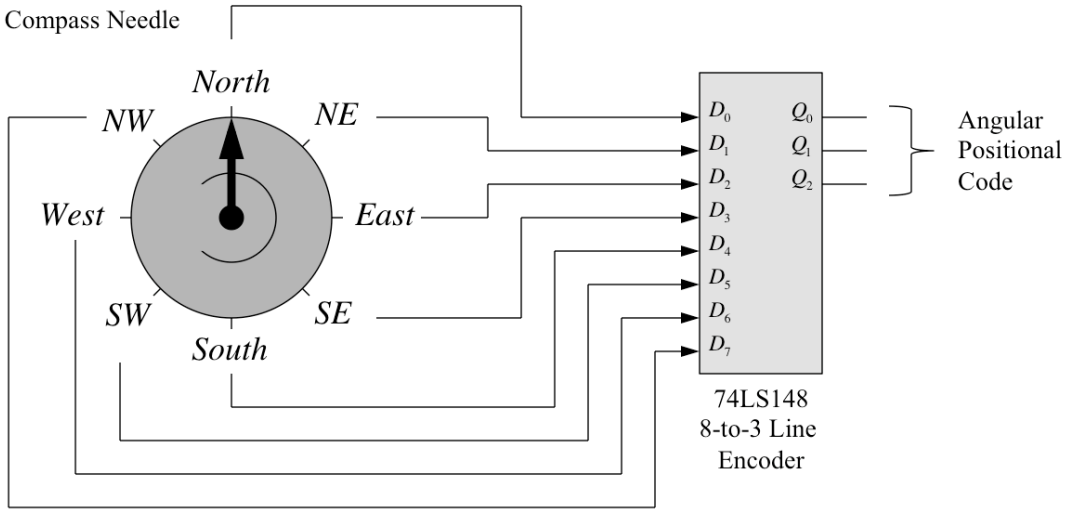


Figure 4: Priority Encoder Navigation

PART A – OBJECTIVES: 20 MARKS

1. Which of the following functions can be used to detect how many empty slots available in a parking lot?

   A. Encoder
   B. Comparator
   C. Multiplexer
   D. Parity Generator

3. Which statement about DEMUX is FALSE?

   A. It is also known as data selector.
   B. A decoder can also be used as a DEMUX.
   C. A DEMUX reverses the function of a MUX.
   D. It takes digital information from one line and distributes it to a given number of output lines.

4. Choose the TRUE statement regarding parity checker.

   A. Parity checker is used to correct an error in the received data.
   B. Parity checker is used to determine numbers of bit 1 in the data.
   C. OR gate is used in parity checker to check for inequality.
   D. Parity checker is produced using parallel adder.

6. If the data select lines of the MUX in Figure 1 are $S_1S_0 = 11$, what would the output be?

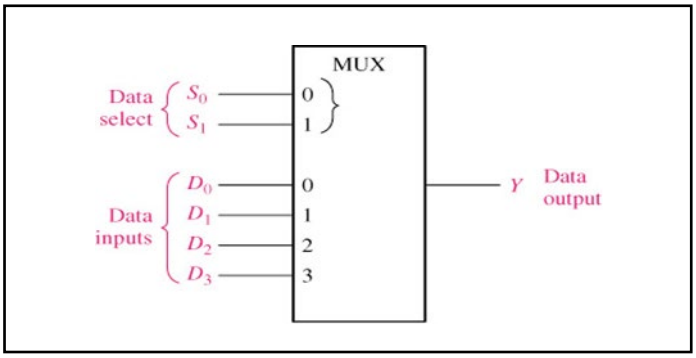


Figure 1

A. LOW

B. HIGH

C. Equal to D_0

D. Equal to D_3

PART B – SUBJECTIVES: 80 MARKS

Assume MSB is the left most bit for all questions.

QUESTION 1 [15 MARKS]

a) Draw a block diagram of 3-bit adder, using both half and full adders. Show clearly all the inputs and outputs in your diagram when input A = 101 and B = 011 are added. [*6 marks*]

b) Draw a logic gate function that would test 2-bit numbers for equality. Name the function. [*4 marks*]

c) Build a truth table for a three inputs decoder with active-high outputs. However, if the value of the input corresponds to 5 or 7, then all output lines should signal an error (bit 1). [*5 marks*]

PART A [20 MARKS]

INSTRUCTION: Write you answer in answer booklet.

1. What is the comparator output for inputs shown in Figure 1?

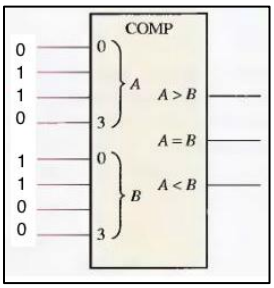


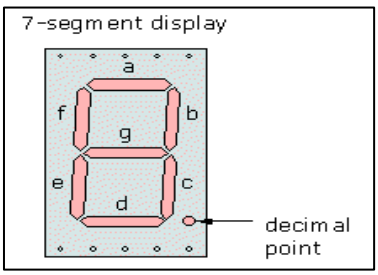
Figure 1

A. Output pin A>B is equal 1

B. Output pin A=B is equal 1

C. Output pin A<B is equal 1

D. All outputs are disabled



3. Figure 3 shows Highest Priority BCD Decoder (HPRI/BCD) IC. What is the binary output if the input pin 2, 3, 5 and 8 are pressed simultaneously?

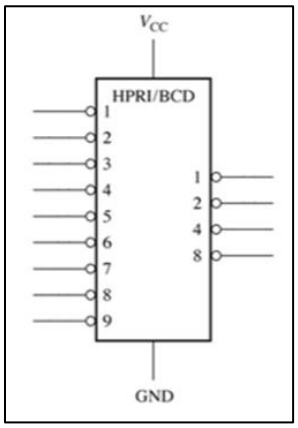


Figure 3

A. 0010
B. 0011
C. 0101
D. 1000

5. Table 1 represents the truth table for which latch?

Table 1

Input		Output
X	Y	Q
0	0	Invalid
0	1	1
1	0	0
1	1	No Change

A. Active-HIGH S-R latch
B. Active-LOW S-R latch
C. Gated S-R latch
D. Gated D latch

6. Flip-flops are normally used for the following applications, **except**:

A. Counter
B. Analog conversion
C. Frequency division
D. Data storage

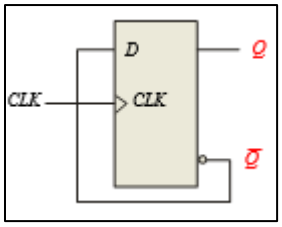


Figure 4

PART B [80 MARKS]

INSTRUCTION: Answer all questions in the answer booklet EXCEPT questions with timing diagram in Q2b. You must show all work to receive credit.

QUESTION 1 [20 MARKS]

a) Draw a schematic block diagram of three (3) bit Parallel Adder using Full Adders. Perform the following arithmetic 3-bit addition shown in Figure 6 and show all the values for this binary additional in your drawing. [7m]

$$\begin{array}{rccc} & 1 & 0 & 1 \\ + & 0 & 1 & 1 \\ \hline & & & \end{array}$$

Figure 6

b) A Bin/Dec decoder is used to activate a series of devices by a microcontroller as shown in Figure 7. Answer the following questions. [7m]

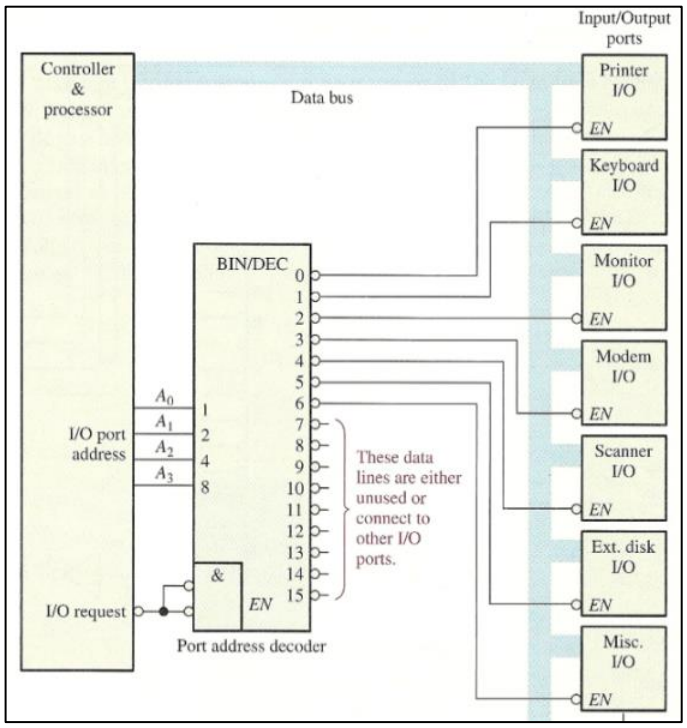


Figure 7

i) Is the Bin/Dec decoder active HIGH or active LOW? What does that mean?

ii) Which device will be activated when the input decoder is 0011?

iii) What is the input decoder when External Disk I/O is activated?

iv) If Input/output is selected as Monitor I/O, draw a simple decoder circuit using inverter and NAND gate.

c) Four different inputs from source *I* are sent to four different output destination *Y* using Mux and Demux as shown in Figure 8. Based on the input of I_3, I_2, I_1, and I_0 given in Figure 8, and inputs S_1, S_0 and M_1, M_0 in Table 3, give the binary answer for *G* and output Demux of Y_3, Y_2, Y_1, and Y_0. [6m]

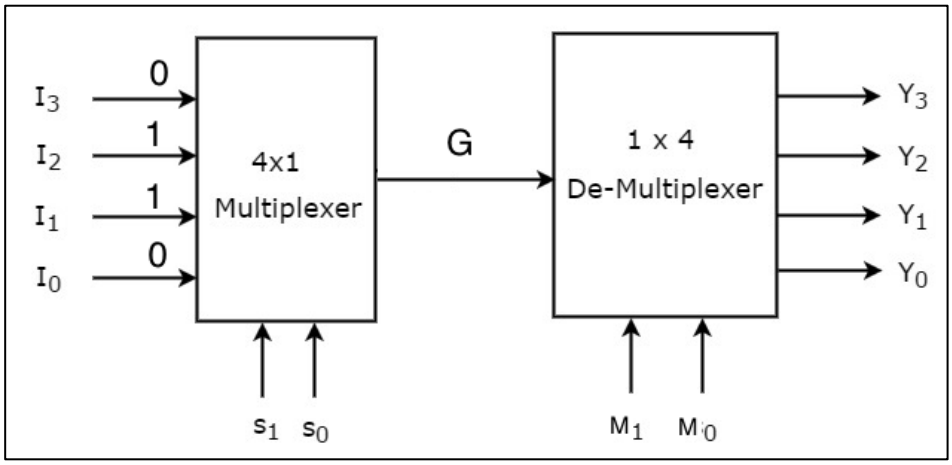


Figure 8

Table 3

S_1 S_0	G	M_1M_0	$Y_3Y_2Y_1Y_0$
01	(i)	10	(iv)
11	(ii)	01	(v)
10	(iii)	00	(vi)

PART A: OBJECTIVES

Answer ALL questions in answer booklet.

1. What is a multiplexer?
   A. It is a type of decoder which decodes several inputs and gives one output.
   B. A multiplexer is a device that has many inputs and a single output.
   C. It takes one input and results into many outputs.
   D. It is a type of encoder which decodes several inputs and gives one output.

4. In a multiplexer the output depends on its _____.
   A. data outputs
   B. select inputs
   C. select outputs
   D. rotate pin

5. Parity generators/checkers use which modulo summation?

   A. Modulo 8

   B. Modulo 16

   C. Modulo 2

   D. Modulo 10

6. A code converter is a logic circuit that _____.

   A. Inverts the given input

   B. Converts into decimal number

   C. Converts data of one type into another type

   D. Converts to hexadecimal

PART B: STRUCTURED
Answer ALL questions in question and answer booklet.

Question 1 [16 Marks]

a) Draw the block diagram for a 3-bit parallel adder using full adder and label clearly the block diagram including the input, sum and carry. Perform addition $011_2 + 110_2$ **,** show the value for each label on the block diagram. [4]

$$\begin{array}{rcccc} & A_3 & A_2 & A_1 \\ + & B_3 & B_2 & B_1 \\ \hline \sum_4 & \sum_3 & \sum_2 & \sum_1 \end{array}$$

b) For set of binary numbers $A_3A_2A_1A_0 = 0010$ $B_3B_2B_1B_0 = 0101$ are assigned to comparator in Figure 1.

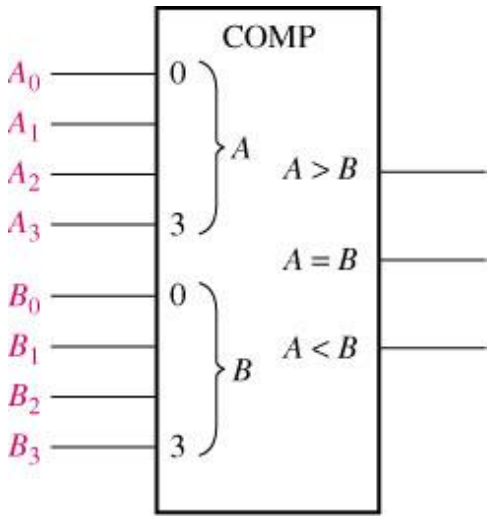


Figure 1

i. Based on Figure 1, explain how comparator works until final output results appear. [3]

ii. Determine the final output result for the comparator in Figure 1. [1]

c) Given an active HIGH 7-segment display in Figure 2 showing a digital number from a decoder that is based on the BCD value, **complete Table 1 in your question booklet APPENDIX A**. [4]

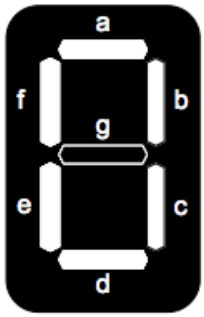


Figure 2

d) Figure 3 in APPENDIX A shows DEMUX connection with outputs labelled as Y_3 Y_2 Y_1 Y_0 and B is the input of DEMUX. Select bit for DEMUX are noted by C_1 and C_0 (C_1 is the MSB). **Draw the waveform for the DEMUX output based on the given values in question booklet Figure 3 APPENDIX A.** [4]